

VIT BHOPAL UNIVERSITY

B.Tech – Computer Science and Engineering

CSE1021

Introduction to Problem Solving and Programming

PROJECT REPORT

Hostel Complaint & Maintenance Management System

Submitted by

Vaibhav Hansraj

Registration No.: 26BCE10988

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1. Introduction

The Hostel Complaint & Maintenance Management System is a Python-based command-line application designed to help hostel students register and manage maintenance complaints. It allows users to register complaints, view records, search complaints, update status, and view basic statistics.

The project applies CSE1021 concepts including variables, data types, input/output, conditions, loops, functions, lists, dictionaries, searching, file handling, and modular programming.

2. Problem Statement

Hostel students may face electrical faults, plumbing issues, cleaning problems, and damaged furniture. When complaints are handled manually, it can become difficult to track their status and find previous complaints. This project provides a simple digital system for registering, searching, tracking, and managing hostel maintenance complaints.

3. Objectives

- Register hostel complaints digitally.
- Store complaint details in an organized way.
- Search complaints easily.
- Track complaint status.
- Display basic complaint statistics.
- Reduce difficulty in managing hostel maintenance complaints.

4. Target Users and Scope

Target Users: Hostel students and hostel maintenance staff.

Scope: The system covers complaint registration, viewing, searching, status updates, and statistics. It does not include online payment, real-time staff tracking, or external notification services.

5. Functional Requirements

ID	Requirement	Description
FR1	Complaint Registration	Allow students to register a hostel complaint.
FR2	Complaint Viewing	Allow users to view all registered complaints.
FR3	Complaint Search	Search by ID, room, type, status, or priority.
FR4	Status Management	Update status to Pending, In Progress, or Resolved.
FR5	Complaint Statistics	Display totals, status counts, resolution rate, type counts, and most common type.
FR6	Data Storage	Store complaint information in a JSON file.

6. Non-Functional Requirements

- Usability – simple command-line interface.
- Reliability – correct storage and validation of complaint data.
- Maintainability – separate Python modules for different functions.
- Error Handling – messages for invalid choices or information.
- Performance – normal complaint records should be processed quickly.

- Data Integrity – stored data remains available after closing and reopening the application.

7. System Architecture

The system follows a simple modular architecture. `main.py` controls the menu. Separate modules handle registration, viewing, searching, status management, statistics, validation, and storage.

User	
↓	
<code>main.py</code>	Main program and menu
<code>registration.py</code>	Register complaint
<code>complaint.py</code>	Generate ID and view complaints
<code>search.py</code>	Search complaints
<code>status.py</code>	Update status
<code>statistics.py</code>	Calculate statistics
<code>validation.py</code>	Validate input
<code>storage.py</code>	Read/write JSON
↓	
<code>data/complaints.json</code>	Complaint storage

8. System Workflow

- Start application → display main menu.
- Register: enter details → validate → generate ID → save.
- View: read stored data → display complaints.
- Search: choose criteria → search stored records → display matches.
- Update status: enter ID → select status → save update.
- Statistics: read data → calculate values → display results.
- Return to menu or exit.

9. Complaint Data Design

Each complaint contains Complaint ID, Student Name, Room Number, Complaint Type, Description, Priority, and Status.

Types: Electrical, Plumbing, Cleaning, Furniture, Other. Priority: Low, Medium, High. Status flow: Pending → In Progress → Resolved.

10. Implementation

10.1 Main Module

`main.py` displays the menu, accepts the user's choice, and calls the appropriate module.

10.2 Complaint Registration

`registration.py` collects details, validates them, generates a unique ID, and saves the record.

10.3 Viewing Complaints

complaint.py loads stored records, generates IDs, and displays complaints.

10.4 Searching

search.py searches by ID, room, type, status, or priority using sequential searching.

10.5 Status Management

status.py updates a complaint to Pending, In Progress, or Resolved and saves the change.

10.6 Statistics

statistics.py calculates totals, status counts, resolution rate, type counts, and the most common type.

10.7 Validation and Storage

validation.py checks required fields and allowed values. storage.py reads and writes the JSON file.

11. Technologies and Tools

- Python 3
- JSON file storage
- Visual Studio Code
- Git
- GitHub

12. CSE1021 Concepts Used

- Variables and data types
- Input/output
- Conditional statements
- Loops
- Functions
- Lists
- Dictionaries
- Searching
- Counting and data processing
- File handling
- Modular programming

13. Testing

Test	Expected Result	Result
Register complaint	Complaint saved with unique ID	Passed
View complaints	Stored complaints displayed	Passed
Search by ID/room/type/status/priority	Matching complaints displayed	Passed
Update status	Status updated and saved	Passed

View statistics	Correct statistics displayed	Passed
Invalid menu choice	Error message displayed	Passed
Invalid type or priority	Input rejected	Passed
Invalid complaint ID	Error message displayed	Passed

Integration testing used the workflow Register Complaint → View Complaint → Search Complaint → Update Status → View Statistics. The complete system test was successful.

14. Screenshots / Results

The project includes screenshots demonstrating the main menu, complaint registration, and complaint statistics. These screenshots are maintained in the project's docs folder in the GitHub repository.

15. Design Decisions

- Python was selected because it is simple and suitable for CSE1021 concepts.
- A command-line interface was selected for a simple first-semester project.
- The modular structure separates different functions and improves maintainability.
- JSON is sufficient for a small complaint-management dataset.
- The Pending → In Progress → Resolved flow provides simple status tracking.
- Input validation prevents incomplete or invalid complaint information from being stored.
- Sequential searching is suitable for a small dataset.
- Statistics use loops, conditions, dictionaries, and arithmetic.

16. Challenges Faced

- Organizing the program into separate modules.
- Validating user input.
- Maintaining data between program runs using JSON.
- Generating unique complaint IDs.
- Testing the complete integrated workflow.

17. Learnings

- Breaking a programming problem into smaller modules.
- Using functions, loops, conditions, lists, dictionaries, and file handling.
- Storing and retrieving structured data using JSON.
- Basic searching and data processing.
- Using Git and GitHub for version control.
- Importance of testing and documentation.

18. Future Enhancements

- Graphical user interface.
- User login and role-based access.
- Real-time maintenance staff tracking.

- Notification services.
- Database storage for larger datasets.
- Complaint timestamps and detailed reporting.

19. GitHub and Project Structure

The project is maintained using Git and GitHub. The repository contains Python source modules, JSON data, documentation, and design diagrams.

Source: main.py, registration.py, complaint.py, search.py, status.py, statistics.py, validation.py, storage.py.

Documentation: README.md, statement.md, requirements.md, architecture.md, workflow.md, design.md, testing.md.

Data: data/complaints.json.

20. Conclusion

The Hostel Complaint & Maintenance Management System provides a simple digital approach for registering, searching, tracking, and analyzing hostel maintenance complaints. It demonstrates the problem-solving and programming concepts covered in CSE1021 and applies them to a practical hostel-related problem.

The completed system is modular, uses JSON for persistent storage, validates user input, supports complaint status tracking, and provides basic statistics. The project also includes documentation, testing, diagrams, and GitHub version control.

21. References

- CSE1021 Introduction to Problem Solving and Programming course syllabus.
- VITYarthi project instructions and evaluation requirements provided for the course.
- Python 3 documentation and language concepts used in the implementation.